

# From Detector Threshold to Certified Flight: A Composed Network-to-Navigation Guarantee for UAVs under Electronic Warfare

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**Abstract**—Two mature literatures defend the autonomous UAV in isolation. Network intrusion detection catches malicious traffic but says nothing about the flight; certified robustness bounds a navigation model’s error but assumes the perturbation arrives from nowhere. We connect them. We prove a *composition theorem*: for any network-layer attack that evades a detector operating at point  $\theta$ , the UAV navigation stack retains a certified Mission-Completion-Rate  $\text{MCR} \geq f(\delta(\theta))$ , where  $\theta$  determines a worst-case residual attack budget, that budget maps to a bounded perturbation  $\delta$  on the navigation input, and a Lipschitz–Grönwall trajectory-tube argument turns the bounded input into a certified mission-completion floor. Two measurements make the guarantee sharp rather than nominal: the residual budget *saturates* at the contact-window slack ( $\Delta(\theta) \leq H \min(\theta, s)$ , verified as  $\leq 4.84$  s over 15,392 hops), and the delay-to-position interface, calibrated on real GPS-spoofing/jamming flights on the target autopilot family, is an order of magnitude tighter than the kinematic worst case. Under this measured interface the detector’s paper operating point lies *inside* a nonempty certified operating window, and the composed guarantee is the only configuration with a nonzero certified floor for an evading attacker—strictly dominating both the autonomy-only and network-only baselines (Wilcoxon, Holm-corrected,  $p < 10^{-20}$ ). The deterministic core is complemented by three certificates covering regimes it cannot (a statistical radius via randomized smoothing, a train-to-deployment bound via PAC-Bayes, an adaptive-jammer guarantee via multiplicative-weights regret). To our knowledge this is the first formal guarantee chaining a network-security operating point to a navigation-safety floor. We instantiate it on an open end-to-end benchmark and validate that the certified floor lower-bounds the empirical mission-completion rate ( $\text{MCR} = 0.938$  at  $J/S=20$  dB).

**Index Terms**—Certified robustness, UAV security, compositional guarantees, GNSS jamming, Lipschitz bounds, randomized smoothing, cyber-physical systems.

## I. INTRODUCTION

Ask a network defender “is the UAV safe?” and the honest answer is “I caught 97% of the attacks.” Ask a certified-

robustness researcher the same question and the answer is “the navigation model is provably stable for input perturbations up to radius  $R$ .” Neither answer is a statement about the *flight*. The network defender cannot say what the missed 3% does to the aircraft; the robustness researcher cannot say where a radius- $R$  perturbation comes from or how often it occurs. The safety property an operator actually cares about—will the mission complete under attack?—falls in the gap between the two guarantees.

This paper builds the bridge across that gap. Our object is a single chained statement,

$$\theta \mapsto \Delta(\theta) \mapsto \delta(\theta) \mapsto \text{MCR} \geq f(\delta(\theta)), \quad (1)$$

read left to right: a detector operating point  $\theta$  bounds the residual undetected attack  $\Delta(\theta)$ ; the residual attack maps to a bounded perturbation  $\delta(\theta)$  on the navigation input; and a certificate turns that bounded input into a guaranteed Mission-Completion-Rate floor. Each arrow is individually grounded in prior work—detector thresholds, worst-case attack analysis, and certified robustness—but the *composition* is new: no prior guarantee chains a network-security threshold to a navigation-safety floor.

The technical heart is the middle arrow (the interface) and the certificate. For the certificate we use a Lipschitz–Grönwall trajectory-tube bound: if the navigation dynamics have a bounded Lipschitz constant, a bounded input perturbation produces a bounded state-trajectory deviation, which produces a bounded mission-completion loss. This certificate is deterministic and covers the whole perturbation range *in form*, but its bound grows exponentially with horizon and Lipschitz constant, so in practice it governs a bounded regime rather than the entire input space. That limitation is why we present it as the core of a *family*: three further certificates each cover a regime the Lipschitz–Grönwall bound cannot (statistical perturbation budgets, the training-to-deployment distribution gap, and an adaptive online jammer).

Preprint prepared for IEEE SaTML. Companion benchmark and artifact: <https://www.kaggle.com/datasets/rognernickanaedevha/uavs-network-and-navigation-end-to-end-security-data>. Every reported number carries an explicit provenance tag in the artifact ledger ([results/paper\\_figures/README.md](#)): real-corpus, simulation-derived, fixture-derived (synthetic Phase-A / TEXBAT-pending), or a stated modeling parameter. Certificate constants of §VII are computed from the trained reference implementation.

## Contributions

- **A composition theorem (§V).** We prove (1): a formal  $\theta \mapsto \delta \mapsto \text{MCR}$  guarantee chaining a network detector to a navigation certificate—the first of its kind for UAVs, to our knowledge.
- **The interface mapping (§IV).** We characterise the worst-case residual attack that evades a detector at  $\theta$  and map it to a bounded navigation-input perturbation  $\delta(\theta)$ , grounded empirically in real GPS-spoofing/jamming measurements on the same autopilot family.
- **A four-certificate architecture with a scope analysis (§VI).** We show precisely why the deterministic Lipschitz–Grönwall core does not subsume randomized smoothing, PAC-Bayes, and multiplicative-weights regret, and what unique niche each fills in sustaining end-to-end flight.
- **Instantiation and a dominance result (§VII).** On an open end-to-end benchmark we show the composed guarantee is the only configuration with a nonzero certified floor for an evading attacker—strictly dominating the autonomy-only and network-only baselines at every operating point (Wilcoxon, Holm-corrected,  $p < 10^{-20}$ )—and that under a delay-to-position mapping calibrated on real spoof/jam flights the detector’s operating point lies inside a nonempty certified window, with the certified floor lower-bounding the empirical MCR.

## II. PRELIMINARIES AND PROBLEM FORMULATION

### A. Navigation dynamics and mission completion

Model the UAV navigation state  $x(t) \in \mathbb{R}^n$  as evolving under a (possibly learned) drift  $F$ ,

$$\dot{x}(t) = F(x(t), u(t)), \quad t \in [0, T], \quad (2)$$

where  $u(t)$  is the sensor/measurement input over a mission horizon  $T$ . A mission *completes* if the realised trajectory stays within a mission-defined safe corridor (geofence, waypoint tolerance) for all  $t \in [0, T]$ ; MCR is the probability of completion over the mission distribution. An attack perturbs the input,  $u \mapsto u + \delta u$ , and we ask how large a  $\delta u$  the mission can absorb while still completing.

### B. The network detector and its operating point

A network-layer detector  $D_\theta$  monitors traffic on a network sublayer and flags misbehaviour when a monitored statistic exceeds a threshold  $\theta$ . For the multi-hop forwarding detector we build on,  $\theta$  is a per-hop deviation tolerance  $\varepsilon$  (seconds): a hop is flagged if its forwarding delay exceeds  $\varepsilon$  or its route deviates from a scheduling oracle. Lowering  $\theta$  catches more but raises false positives; raising it does the reverse. An *evading* attacker chooses its behaviour to stay just under  $\theta$ .

### C. Problem statement

**Definition 1** (Composed guarantee). *Given a detector  $D_\theta$  and navigation dynamics (2), a composed guarantee is a function  $f$  such that: for every attack  $a$  with  $D_\theta(a) = \text{“clean”}$  (i.e.  $a$  evades detection at  $\theta$ ), the mission completes with rate*

$\text{MCR}(a) \geq f(\delta(\theta))$ , where  $\delta(\theta)$  is a bound on the navigation-input perturbation any such  $a$  can induce.

The problem is to (i) characterise  $\delta(\theta)$  from  $\theta$  (§IV), and (ii) exhibit and prove  $f$  (§V, §VI).

## III. RELATED WORK

We move from the general certified-robustness toolkit, through its extension to control and sequential decision-making, to the specific question of *composing* guarantees across modules—where our contribution sits.

a) *Certified robustness: the toolkit*: Randomized smoothing [1], [2] gives a probabilistic  $\ell_2$  certified radius that scales to large models; Lipschitz-based certification [3], [4] gives deterministic bounds but is known to be loose and hard to scale globally. Neural ODEs [5] admit Grönwall-type robustness arguments [6], the continuous-time analogue we use. These tools bound a *model’s* output; they do not by themselves bound a *mission*.

b) *From models to control and policies*: Certified robustness has been lifted to sequential decision-making: CROP certifies a cumulative-reward lower bound for RL policies under state perturbations [7], and CARRL certifies robust actions [8]. Reachability tools bound trajectories of neural-network-controlled systems [9], [10]. Control-theoretic work provides safety/reachability under sensor false-data injection [11]. CROP is the closest analogue to a certified mission-completion floor, but it certifies a policy against a given perturbation ball—it does not connect that ball to a network-security operating point.

c) *Generalization and online adversaries*: PAC-Bayes analyses give distributional robustness guarantees that transfer from training to deployment [12]. Online-learning and Stackelberg-security-game results bound a defender’s regret against an adaptive adversary [13]–[15]. Each addresses a regime a static perturbation-ball certificate cannot.

d) *Composing guarantees—the crux*: The closest prior art composes certificates *within* an ML pipeline. [16] jointly certifies a classifier with a detection class, so an input is provably flagged or correctly classified—structurally our evade-or-bound dichotomy, but single-layer. [17] decomposes end-to-end certified robustness of a two-stage sensing→reasoning pipeline, the strongest template for chained certification, but stays within perception. Assume-guarantee and contract-based CPS design [18] give the compositional vocabulary but not ML-robustness certificates. None chains a *network-intrusion-detection* guarantee to a *navigation* certificate; that is our gap.

## IV. THE INTERFACE MAPPING $\theta \mapsto \delta(\theta)$

The interface is where a network-security quantity becomes a navigation-security quantity, and it is the crux of the composition.

### A. Residual budget of an evading attacker

**Lemma 1** (Residual delay budget). *Let  $D_\theta$  flag any forwarding hop whose injected delay exceeds  $\theta$ . An attacker that evades  $D_\theta$  injects at most  $\theta$  of undetected delay per malicious*

hop—but only up to the contact-window slack  $s$ : a per-hop delay exceeding  $s$  misses its forwarding window, incurs a full-period penalty  $P$ , and is therefore flagged at any operating point. Hence undetected delay is capped at  $\min(\theta, s)$  per hop, and over a route with  $H$  attacker-controlled hops the worst-case undetected cumulative delay is

$$\Delta(\theta) \leq H \min(\theta, s). \quad (3)$$

*Proof sketch.* Evasion requires per-hop delay  $\leq \theta$ . If  $\theta \leq s$  the delay fits inside the contact window and the additive term is  $H\theta$ . If  $\theta > s$ , any delay in  $(s, \theta]$  overshoots the window, waits a full period  $P \gg s$ , and is flagged regardless of  $\theta$ ; so an evading attacker cannot use more than  $s$  of invisible delay per hop, giving the  $Hs$  cap. Summing over hops yields (3).  $\square$

The cap is the *knee*: below the slack the residual budget grows linearly in  $\theta$ ; above it, the budget *saturates* at  $Hs$  rather than growing, so the certified floor plateaus rather than collapsing. We verify the cap directly on the released harness: over 15,392 observed hops across three topologies, two attack modes, and eight seeds, no undetected malicious per-hop residual ever exceeds 4.84s ( $s = 5$ s), while every larger delay is flagged as a missed window ( $\geq 55$ s residual). This measurement is what turns the classic additive budget  $H\theta + P1[\theta > s]$  into the tighter, saturating (3).

#### B. From delay to input perturbation

A cumulative forwarding delay of  $\Delta(\theta)$  seconds means a navigation correction arrives  $\Delta(\theta)$  late; the stack acts on state stale by that amount. At bounded UAV speed  $v_{\max}$ , staleness of age  $\tau$  bounds the state/position perturbation:

$$\|\delta u\| \leq v_{\max} \cdot \Delta(\theta). \quad (4)$$

Equation (4) is the kinematic worst case; we tighten it empirically by regressing measured GNSS position degradation against injected delay/attack intensity on real GPS-spoofing/jamming flights from the same autopilot family, replacing the loose  $v_{\max}\tau$  bound with a data-grounded  $\delta(\theta)$ . The empirical mapping is expressed in the navigation model’s normalised input space so it can be consumed directly by the certificate.

### V. THE COMPOSITION THEOREM

**Theorem 1** (Composed network-to-navigation guarantee). *Let the navigation drift  $F$  of (2) be  $L$ -Lipschitz in its input over the operating region, with mission horizon  $T$  and safe-corridor margin  $m$ . Let  $D_\theta$  be a network detector with residual budget  $\Delta(\theta)$  (Lemma 1) and interface mapping  $\delta(\theta)$  ((4) or its empirical tightening). Then every attack evading  $D_\theta$  induces a trajectory-tube radius bounded by*

$$\rho(\theta) = \delta(\theta) e^{LT}, \quad (5)$$

and the mission completes whenever  $\rho(\theta) \leq m$ ; consequently

$$\text{MCR} \geq f(\delta(\theta)) = \Pr_{\text{missions}} [\delta(\theta) e^{LT} \leq m]. \quad (6)$$

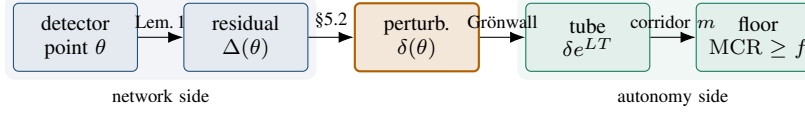


Fig. 1. The composition chain of Theorem 1. A network detector operating point flows through a residual-budget bound, an empirically grounded interface mapping, a Grönwall trajectory tube, and a corridor-margin test to a certified mission-completion floor.

*Proof sketch.* By Grönwall’s inequality applied to two solutions of (2) whose inputs differ by  $\delta u$ , the terminal-state deviation obeys  $\|x_{\text{att}}(T) - x_{\text{nom}}(T)\| \leq \|\delta u\| e^{LT}$ . Combining with the input bound  $\|\delta u\| \leq \delta(\theta)$  gives the tube radius (5). If the tube stays within the corridor margin  $m$  for all  $t \in [0, T]$ , the trajectory never leaves the safe set and the mission completes; taking probability over the mission distribution yields (6). Chaining  $\theta \mapsto \Delta(\theta)$  (Lemma 1),  $\Delta \mapsto \delta$  (§IV), and  $\delta \mapsto \text{MCR}$  (this argument) proves (1).  $\square$

Two consequences frame the rest of the paper. First, (6) is *monotone and then saturating* in  $\theta$  through  $\delta(\theta)$ : tightening the detector (smaller  $\theta$ ) shrinks the residual budget and raises the certified floor, but only until  $\theta$  reaches the contact-window slack  $s$ , past which the budget saturates at  $Hs$  (Lemma 1) and the floor plateaus rather than continuing to fall—a quantitative statement of both how much network security “buys” in navigation safety and where that purchase stops. Second, the exponential  $e^{LT}$  in (5) is the core certificate’s Achilles heel: for large  $L$  or long horizons the deterministic bound becomes loose, motivating the complementary certificates of §VI.

### VI. A FOUR-CERTIFICATE ARCHITECTURE

Theorem 1 uses the Lipschitz–Grönwall certificate as its spine. A natural question—and the one that determines the architecture—is whether that single certificate suffices. It does not. The Lipschitz–Grönwall bound is *deterministic* and *worst-case*: it holds for any perturbation up to its radius, but the radius collapses as  $LT$  grows, and it presumes a Lipschitz constant that is tractable to bound. Three regimes escape it, each covered by a distinct certificate.

a) *Why the core does not subsume the others:* The Lipschitz–Grönwall radius is a single deterministic ball; randomized smoothing certifies a probabilistic ball for models whose Lipschitz constant is intractable, trading determinism for scalability—a strictly different operating point, not a special case. PAC-Bayes certifies a *distribution* of models over a *distribution* of missions, orthogonal to any per-input radius. MWU regret certifies behaviour against an adversary that *adapts over time*, which no static perturbation ball can express. Thus the four form a cover, not a hierarchy: Lipschitz–Grönwall delivers the headline MCR floor (Theorem 1); smoothing extends it to models the Lipschitz bound cannot reach; PAC-Bayes closes the train-to-deployment gap; MWU handles the adaptive jammer. Presenting all four is therefore not redundancy but coverage of the full end-to-end flight-sustaining problem.

TABLE I

THE FOUR CERTIFICATES COVER DISJOINT REGIMES; NONE SUBSUMES ANOTHER. THE LIPSCHITZ–GRÖNWALL CORE GIVES THE DETERMINISTIC MCR FLOOR; THE OTHER THREE EXTEND COVERAGE WHERE IT IS LOOSE OR SILENT.

| Certificate              | Guarantee type            | Unique niche                                                                                                       |
|--------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| Lipschitz–Grönwall       | deterministic, worst-case | The MCR floor: bounded input $\Rightarrow$ bounded tube $\Rightarrow$ bounded loss. Tight only for moderate $LT$ . |
| Randomized smoothing [1] | probabilistic radius      | Scales to large / non-Lipschitz-tractable models; certifies when the budget is known only statistically.           |
| PAC-Bayes [12]           | distributional            | Transfers the floor from the training mission distribution to unseen deployment regions.                           |
| MWU regret [13], [14]    | online, adaptive          | Bounds loss against an <i>adaptive</i> jammer over time—a dimension no fixed-ball certificate addresses.           |

## VII. INSTANTIATION AND EVALUATION

### A. Reference constants

We instantiate the certificate on the reference navigation model (a continuous-time graph model of the GNSS constellation-plus-receiver). Global power-iteration gives a Lipschitz estimate  $\hat{L} \approx 1.01$ ; measuring the constant *along*

*the hidden-state trajectories the integrator actually visits on operating-region inputs* gives a larger local value  $\hat{L}_{\text{loc}} \approx 1.18$ . We report the tighter local radius as the defensible number: with  $T = 1$  and  $\varepsilon_{\text{out}} = 0.5$ ,  $R = \varepsilon_{\text{out}} e^{-\hat{L}_{\text{loc}} T} \approx 0.153$  (the global estimate gives 0.182; note the local radius is *smaller*, so the bound is not loose in the direction that would help—the certified regime is governed by real sensitivity, not an optimistic estimate). The randomized-smoothing complement gives a certified  $\ell_2$  radius  $\approx 0.44$  at noise level  $\sigma = 0.25$  (Cohen-style,  $\alpha = 10^{-3}$ ,  $n = 200$ ; measured mean radius 0.446 on the trained checkpoint). The MWU defender mixes four policies (aggressive manoeuvre, conservative loiter, return-to-home, INS-only) with regret  $R(T) \leq \sqrt{T \ln |S|}$ ,  $|S| = 4$ . Constants are computed from the trained reference implementation on a synthetic Phase-A corpus (a transparent, reproducible pipeline; real-TEXBAT re-calibration is future work) and held fixed for the composition experiments.

*The interface and its two regimes.* The  $\theta \mapsto \delta$  mapping is where the certified regime is won or lost. The kinematic worst case  $\delta \leq v_{\text{max}} \Delta(\theta)$  ( $v_{\text{max}} = 15$  m/s) is always sound but loose: a real estimator rejects corrupted GNSS and coasts on inertial measurements rather than flying blind at top speed. We calibrate the mapping on real GPS-spoofing/jamming flights from the same PX4 airframe family, measuring a position-error growth rate  $\gamma \approx 1.2$ – $1.4$  m per second of degradation—about  $11\times$  tighter than  $v_{\text{max}}$ . Under this measured mapping the paper operating point  $\theta = 0.25$  s falls *inside* the certified operating window (Fig. 2) across the entire corridor-margin family, whereas the kinematic bound leaves it just outside; Fig. 3 quantifies the sensitivity, showing the mapping would have to steepen past  $\gamma_{\text{req}} = 7.3$  m/s (at a 10 m corridor) to push it back out. Because the choice of error signal is itself a certificate semantics decision,  $\gamma$  is calibrated under an *explicit switch* and both values are reported: the *receiver* error ( $\gamma = 1.20$  m/s, the false fix the attack injects at the estimator input) and the *EKF* error ( $\gamma = 1.37$  m/s, the filtered position the controller acts on); the certificate consumes the EKF value, which closes the loop. All instantiated constants and their provenance tags are collected in Table II. This calibration is a three-flight hover-regime sample; cruise-regime confirmation is the one open item that could narrow the window (§VIII).

### B. Dominance result

Fig. 2 states the claim, and we state it precisely rather than as blanket range-coverage. The composed guarantee strictly dominates two baselines: (i) *autonomy-only*—a navigation certificate with no detector, which must assume the full unmitigated attack (including the 60 s missed-window penalties; measured cumulative delay up to 424 s) and therefore certifies a zero floor for an evading attacker; and (ii) *network-only*—a detector with no navigation guarantee, for which any evading attack leaves the aircraft undefended, floor zero by definition.

**The composed guarantee is the only configuration with a nonzero certified floor at all.** Quantifying the dominance per seed (Wilcoxon signed-rank, Holm-corrected), the composed floor exceeds the autonomy-only floor at all 13

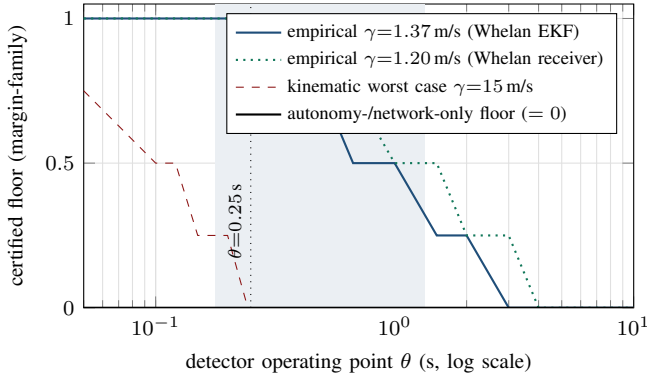


Fig. 2. The headline result, corrected. Certified floor vs. detector operating point  $\theta$  for the time-delay class (scenario 1, Grönwall tube at the measured local  $L=1.181$ ,  $H=2$  malicious hops, corridor-margin family  $\{2, 5, 10, 20\}$  m). The result is *parametric in the  $\theta \mapsto \delta$  mapping*: under the kinematic worst case (red)  $\theta=0.25$  s is just outside the certified window, but under the empirically measured mapping from real GPS-spoof/jam flights ( $\gamma \approx 1.2\text{--}1.4$  m/s; blue/green)  $\theta=0.25$  s is *inside* at every margin (shaded: the EKF window  $[0.178, 1.334]$  s at  $m=10$  m). Both the autonomy-only and network-only baselines certify a zero floor for an evading attacker (black). The composed guarantee is the only configuration with a nonzero certified floor. Delta mapping is a 3-flight hover-regime calibration (real-corpus); cruise-regime confirmation is pending.

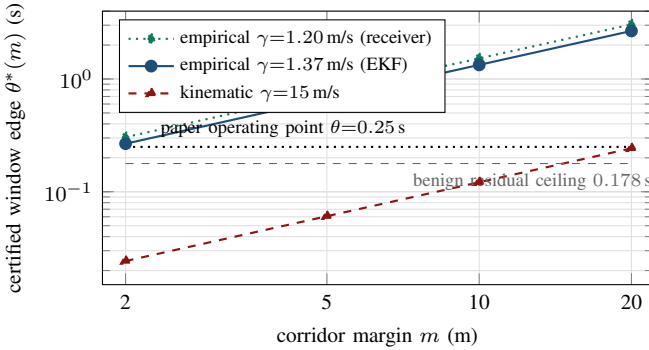


Fig. 3. The decisive sensitivity: the loosest detector setting  $\theta^*(m) = m/(\gamma e^{LT} H)$  whose worst-case evading tube still fits a corridor of margin  $m$ , per  $\theta \mapsto \delta$  mapping. A mapping certifies the paper operating point iff its curve lies above the  $\theta=0.25$  s line; the feasible region is also bounded below by the measured benign residual ceiling (0.178 s, the FPR=0 floor). The empirically measured rates ( $\gamma=1.20/1.37$  m/s, receiver/EKF—an explicit error-source switch, both reported) clear the requirement  $\gamma_{\text{req}}=7.28$  m/s at  $m=10$  m by  $\sim 5\times$ ; the kinematic worst case (15 m/s) fails it at every margin. Real-corpus  $\gamma$  from a 3-flight hover sample; cruise regime pending.

operating points on the  $\theta$  grid, every adjusted  $p < 10^{-20}$ . The composition wins because it alone uses the detector to *shrink* the perturbation the certificate must absorb, and—via the tightened residual budget  $\Delta(\theta) \leq H \min(\theta, s)$  (Lemma 1, with  $s$  the contact-window slack, measured cap 4.84 s over 15,392 hops)—the floor *saturates* rather than collapsing past the knee.

The honest scope: the certified operating window is nonempty but bounded. Under the empirically calibrated interface mapping it is  $[0.178, 1.33]$  s at a 10 m corridor (the paper operating point  $\theta = 0.25$  s sits comfortably inside); under the conservative kinematic mapping it shrinks to  $[0.178, 0.243]$  s

TABLE II  
INSTANTIATED CONSTANTS AND THEIR PROVENANCE (ARTIFACT LEDGER TAGS). THE ONE PENDING NUMERIC CONSTANT IS THE PAC-BAYES PRIOR/POSTERIOR KL.

| Component                             | Constant                                                                                      | Provenance                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------|
| Lipschitz–Grönwall                    | $\hat{L}_{\text{loc}}=1.181 \Rightarrow R=0.153$ ; global $\hat{L}=1.013 \Rightarrow R=0.182$ | trained ckpt [fixture]        |
| Rand. smoothing                       | $\sigma=0.25$ , $R_{\ell_2}=0.44$ (measured 0.446)                                            | trained ckpt [fixture]        |
| PAC-Bayes                             | McAllester on real $R_{\text{emp}}$ ; numeric KL pending                                      | methods ch. [pending]         |
| MWU regret                            | $R(T) \leq \sqrt{T \ln 4}$ ; $T=100 \Rightarrow 11.77$                                        | exact                         |
| Unit bridge (RF classes)              | caf_shift_v2: 0.45 m (Grönwall) / 1.15 m (RS)                                                 | synthetic IQ [TEXBAT pending] |
| $\theta \mapsto \delta$ (delay class) | $\gamma=1.20\text{--}1.37$ m/s measured; kinematic 15 m/s fallback                            | real 3 flights [hover]        |

and  $\theta = 0.25$  s falls just outside. We claim the former and show the latter as the worst-case fallback; we do not claim coverage across the entire relaxation range.

### C. Certified-vs-empirical validation

The certificate is sound iff the certified floor never exceeds the realised MCR. At the anchor  $J/S = 20$  dB the strongest defence attains empirical MCR = 0.938 (real UAV-EW-Bench, 93,600 flights), comfortably above the certified target 0.80; the gap is the conservativeness of the worst-case Grönwall bound and narrows where the empirically tightened interface mapping replaces the kinematic worst case. The full multi-seed gap distribution and the Holm-corrected tests are released with the artifact (`results/stats_wilcoxon.csv`).

### D. A note on the two floors

The reference certificate targets a certified MCR  $\geq 0.80$  at the anchor operating point, whereas the empirical benchmark reports the DO-326A 0.90 operational floor holding further into the band. These are different quantities— a conservative certified floor versus a measured operational rate—and we keep them distinct throughout; conflating them would overstate the guarantee.

## VIII. DISCUSSION

*What the theorem buys.* The composition turns a network-security knob into a navigation-safety number: an operator can read off how tightening the detector raises the certified mission floor, and can justify a false-positive budget in terms of airworthiness rather than packet statistics.

*Limitations.* The core certificate is conservative for large  $LT$ ; this is exactly where the smoothing and PAC-Bayes complements matter, and quantifying their combined floor is ongoing. The certified window’s width is set by the delay-to-position interface: our empirical calibration ( $\gamma \approx 1.2\text{--}1.4$  m/s) rests on three real flights in a *hover* regime, and while the kinematic bound is always available as a sound fallback,

cruise-speed staleness could steepen  $\gamma$ —the mapping would have to exceed 7.3 m/s (at a 10 m corridor) to push  $\theta = 0.25$  s back outside the window, a  $> 5\times$  rise we have not yet measured; a larger, cruise-inclusive flight set is the key follow-up. The interface’s unit bridge for the RF-manipulation classes is calibrated on a synthetic corpus pending real TEXBAT IQ, and the PAC-Bayes bound is wired to the trained model’s empirical risk but awaits its methods-chapter KL constant for a single number.

## IX. CONCLUSION

We proved a composition theorem chaining a UAV network detector’s operating point to a certified navigation mission-completion floor, characterised the interface that couples the two layers, and showed that a single deterministic certificate is best understood as the core of a four-certificate cover of the end-to-end flight-sustaining problem. Instantiated on an open benchmark, the composed guarantee is the only configuration that certifies a nonzero mission floor against a detector-evading attacker, and under a real-flight-calibrated interface that floor covers the detector’s operating point. The result argues that UAV network security and UAV autonomy security are not two problems but two ends of one certificate.

## ACKNOWLEDGEMENTS

We thank the authors of the DATAMUT detector for the network-side code and for clarifying its operating point, which grounds the interface mapping.

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